Analogue Experiment 2: Filter Circuit

Introduction

In this experiment, we use both theoretical analysis and real-world simulation to investigate the behavior of a basic RC (Resistor-Capacitor) low-pass filter circuit. In electrical systems, low-pass filters are crucial because they reduce higher frequencies while allowing signals with frequencies below a certain cutoff to pass. Designing circuits for signal processing, audio engineering, and other filtering applications is made easier by knowing how an RC filter reacts to different input frequencies.

There are several steps to the experiment, starting with theoretical calculations of the filter's impedance and frequency response at specific frequencies. We simulate the circuit and examine the output's amplitude and phase shift in relation to the input using PSPICE software. After that, we experimentally replicate these findings using an oscilloscope and signal generator to compare simulated data with.

Background

An RC low-pass filter consists of a resistor and capacitor in series, with the output taken across the capacitor. The frequency response of this circuit depends on the combined effects of resistance and capacitance, which together determine the cutoff frequency (f_c).

$$fc = \frac{1}{2\pi RC}$$

Frequencies that are below the value of F pass through the circuit with minimal interference whereas values that are above get their altitudes decreased.

The relationship between the input and output voltage in an RC filter depends on the resistor and the capacitor, The equation for this is given as;

$$\frac{V_{out}}{V_{in}} = \frac{1}{\sqrt{1 + (2\pi fRC)^2}}$$

Therefore, as frequency increases there is a higher V_{in} compared to V_{out} hence the voltage drop increases leading to a reduced output.

The filter circuit is often measured using a decibel scale to measure V_{out} / V_{in}. A common reference point is the cutoff frequency, where the output drops to 70.7% or –3 dB, which is the point which marks the transition between passing and decreasing frequencies.

Method

The first step was calculating the ratio of the output to input voltages for both 10kHz and 100kHz and then converting them into their decibel equivalent. The 70.7% (-3dB) frequency was also found.

The next step was setting up the RC circuit with a 10 kΩ resistor and 10nF capacitor on PSPICE. The voltage source was set to be a sin wave with an amplitude of 10v and frequency of 10kHz. The schematic was then configured for transient analysis from 0-1000µs with data stored every 10µs and a maximum step of 5µs, to observe the circuit's output waveform at 10khz. The simulation is then run and the graph of the voltage ratio is produced.

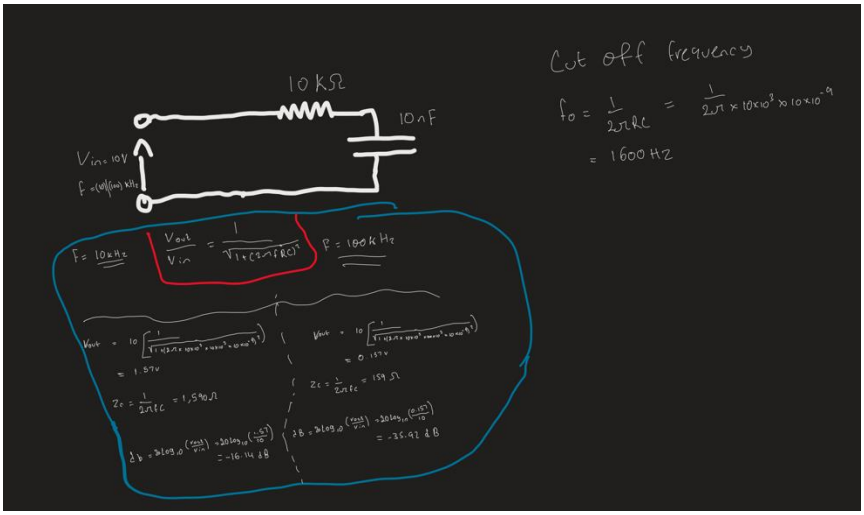
The same is then repeated for the 100kHz but changing the maximum step to 0.5µs

The last schematic that was set up had the sin voltage source changed to an AC source allowing the configuration of an AC sweep from 10Hz to 10MHz with 10 points per frequency decade. The graph is then plotted on a logarithmic scale and the cut off frequency point is identified (-3dB) and the expected slope of 20dB per decade after the cutoff point is displayed.

To accommodate our findings a physical circuit using a waveform generator was set up using the same conditions as the schematic previously simulated, other than the input voltage which was changed to produce a 1v amplitude sine wave at varied frequency's between 1kHz-19kHz and the average input and output voltages were recorded to create a ratio in decibels that was then graphed against the respective frequency.

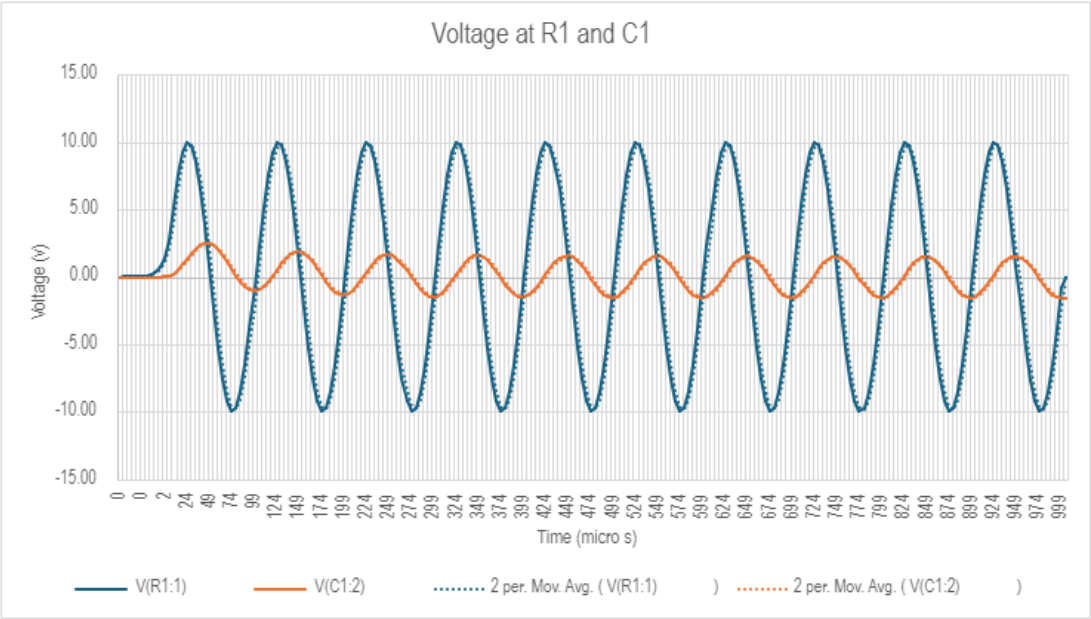
Results

Preliminary Calculations



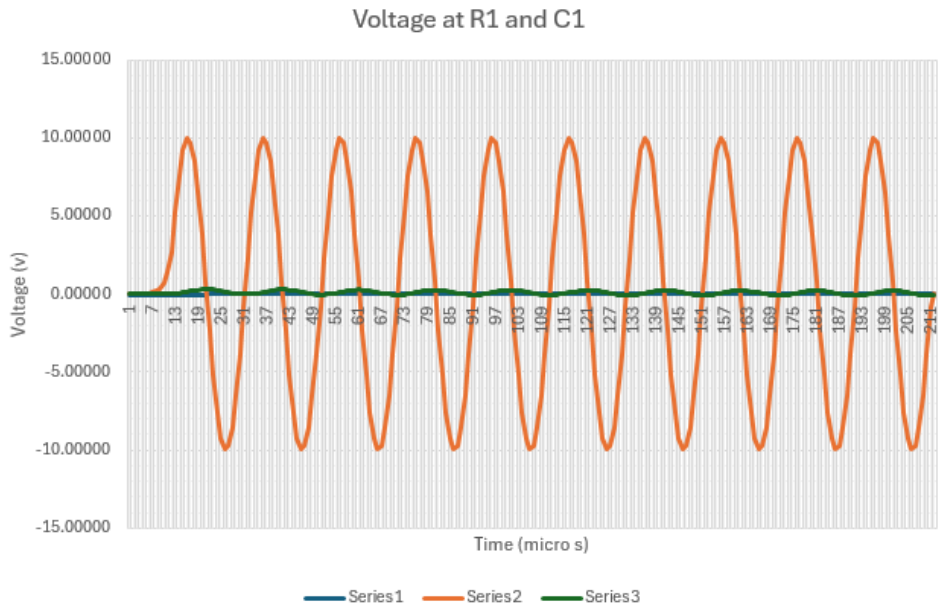
The calculations showed that for 10kHz the impedance was 1,590 Ω and the voltage out was 1.57v. for 100khz the impedance was 159 Ω and the voltage out was 0.157v. The cutoff frrquency was also calculated to be 1600H.

10kHz Graph



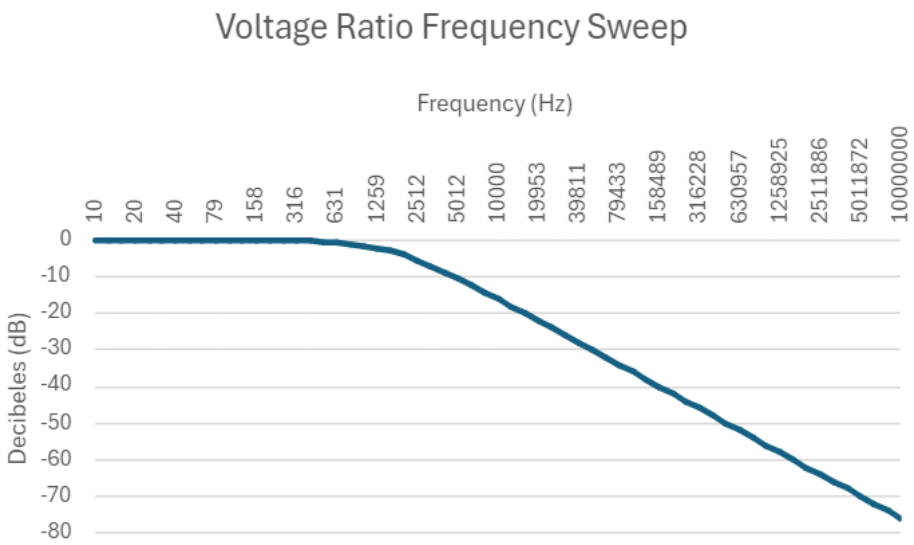
This graph produced by the simulation shows us that the average peak voltage of the output is around our calculated voltage of 1.57v

100kHz Graph



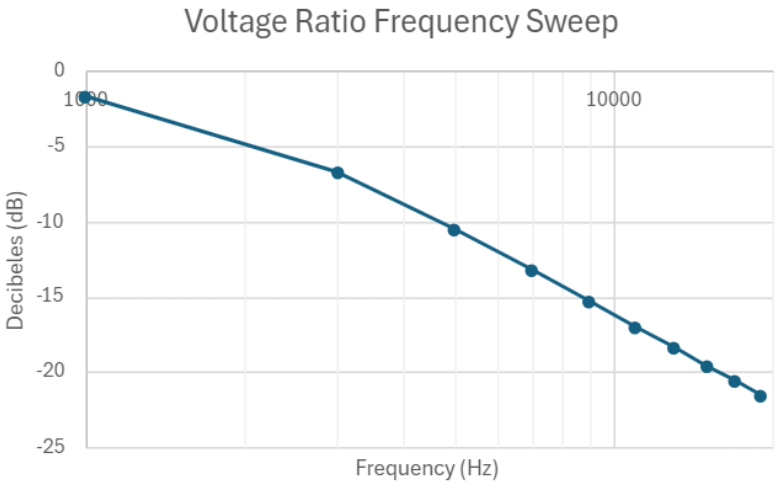
Similarly, this graph produced by the simulation shows us that the average peak voltage of the output is around our calculated voltage of 0.157v

Voltage Ratio Frequency Sweep



Every time the frequency is increased by a factor of 10 the decibels decrease by 20 after the cutoff point of –3dB. This gives a gradient of

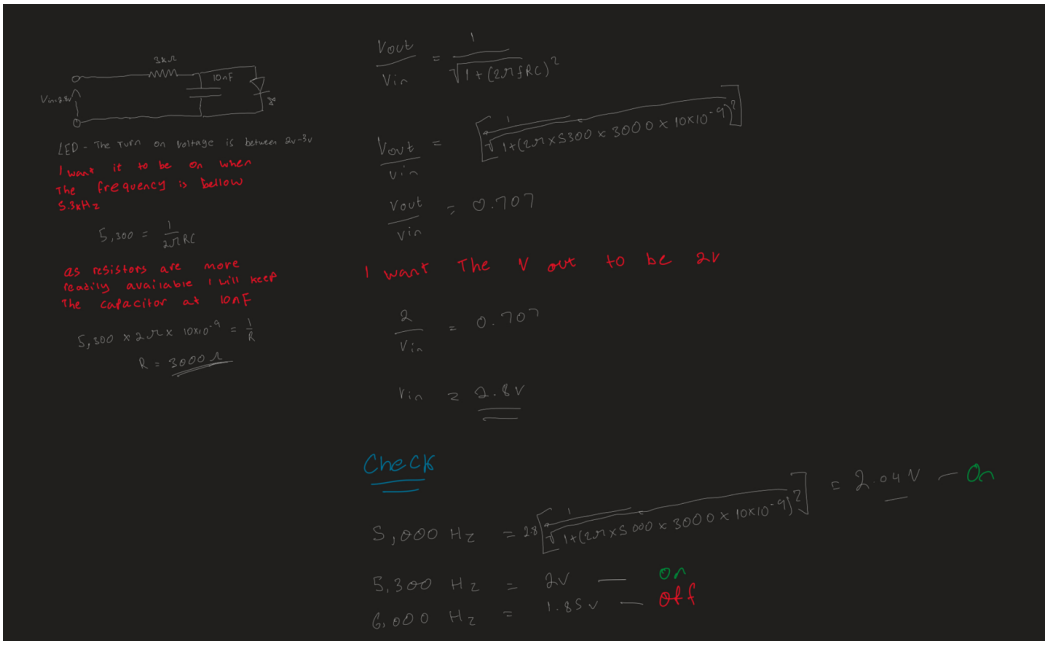
Physical Experiment



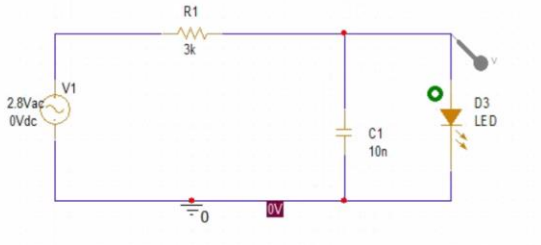
This same experiment is done using a waveform generator and a digital osciliscope and similar properties can be derived as in the previous graph where the cutoff point (though not distinctively) is at the –3dB point and decreases by 16 for every increase in factor of 10 which is close to the accepted value.

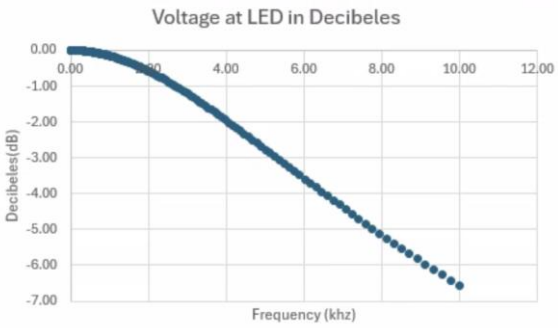
Designed Circuit

The first thing I did was the raw calculation for the different parameters I would need to set my circuit up with.

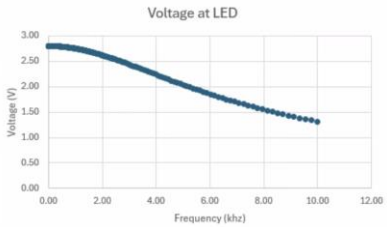


Using these values, I was able to export them into PSPICE and run a simulation





Looking at the graph I got from the simulation we can derive that when there is –3dB we get the cut off voltage that was desired of 5300hz.



Similarly, when we look at the graph of voltage we can see that after the cut off frequency all the values of voltage are below 2 and therefore the LED is off, like originally desired.

Discussion

The RC low-pass filter experiment generated findings that were nearly identical to theoretical expectations, with only small differences. Our experiments revealed a 16 dB loss each decade, when theoretically an RC low-pass filter should suppress high-frequency signals at a rate of 20 dB per decade. This discrepancy may result from measurement restrictions or component tolerances, where actual resistors and capacitors may differ somewhat from their quoted values. At higher frequencies, when excessive inductance and resistances can affect circuit behavior, these minor differences become more noticeable.

Beyond this minor difference, the experiment showed important RC filter properties. Impedance values at 10 kHz and 100 kHz were calculated and measured, and they aligned well, confirming the expected result at various frequencies. The formula;

$$fc = \frac{1}{2\pi RC}$$

was supported by the measured -3 dB cutoff point, which was also close to the expected frequency.

For the circuit I designed a light turned on at low frequency values because the output voltage for the circuit reached the required level for the LED. The practical use of RC filters in systems that need frequency-triggered reactions, such as audio equipment that activates various speakers at specified ranges.

Conclusion

The Lab was useful for applying our theoretical knowledge to real world applications as the majority of the theoretical calculations I completed were directly corelating to the simulations and the physical circuits. Building my own circuit also helped me understand how some applications work in real life and was quite enjoyable as the result I wanted was produced.